

Web Accessibility Barriers (and Solutions)

Millions of people have disabilities that affect their use of the Web. Currently most Web sites and Web software have accessibility barriers that make it difficult or impossible for many people with disabilities to use the Web

<http://www.washington.edu/doit/videos/index.php?vid=35>
<http://www.washington.edu/doit/videos/index.php?vid=35>

Examples of barriers for blind people (using screen readers)

Non-text content

Barrier: screen readers are unable to understand images and graphs (and controls and other structural elements that do not have equivalent text alternatives)
Solution: provide appropriate alternative text (read by screen readers in place of images)

Alternative text 1/5

Alternative text can be presented in two ways:
. Within the alt attribute of the img element (or the longdesc attribute)
. Within the context or surroundings of the image itself.

If the image is either a photograph, drawing or painting, the alt text needs to describe the image in as short of a phrase as possible

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Alternative text 2/5

Images that contain text should generally be coded to just include that text as the alt text.

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Alternative text 3/5

Decorative Images:

many images are used on websites only for visual interest, they aren't meant to convey any meaning or important information. In this case, it is best to use what is called NULL alt text or empty alt text.

Alternative text 4/5

When the image is complex, including things such as charts, data, statistics, etc. where a short description is not possible, the absolute best solution, that would benefit all visitors, would be to include a thorough description of the complex image in the content of the page, immediately before or after the image. If you don't want to add more content to your page, another alternative would be to create another web page with just a description through the longdesc attribute..

Alternative text 5/5

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The longdesc linked file or the page content should then contain a description similar to the following:
The chart below shows trends from 2007 to 2011 comparing the total number of OSU enrolled students to the number of new students each academic year. During the 2007-2008 year there were 19753 students, of which, 5420 were new students, during the 2008-2009...

<http://webaim.org/techniques/alttext/>

Visual layout

Barrier: screen readers cannot realize how a page is organized

Solution: provide a method that allows users to skip navigation or other elements that repeat on every page

Skip.... Links

The link can be hidden through CSS: so it will be perceived only by users of browsers that ignore styles (such as: screen readers)

Skip to main content

Bad links (e.g. Click Here)

Often blind people use the tab key to navigate from link to link

Barrier: screen reader users may choose to read only the links.. but not always they make sense..
Solution: ensure links make sense out of context (avoiding "click here" and "more..")

Bad links and good links

To see the list of our products, [click here](#). (bad)
See the list of [our products](#). (good)

Tables for layout (linearization) (linearization)

Barrier: screen readers essentially ignore the fact that the content is inside of a table and read the content in the order that it appears in the code
Solutions:

- . if the order of the content in the code is logical, then the linearized reading order is logical
- . use CSS instead of tables

How to test for linearization

Disable CSS

Doing this makes everything look like a column of text

Use the WAVE accessibility tool

<http://wave.webaim.org/>

Bad use of Javascript

Blind people can not use the mouse

Barrier: some JavaScript event handlers are device dependent (e.g., they require the use of a mouse)

Solutions:

- . make sure that your page does not rely on mouse to function
- . provide keyboard-based alternatives (onFocus instead of onMouseover)

Visual layout

Barrier: visual content in web videos and animations may provide important information

Solution: provide a transcript

Forms

Barrier: bad code or organization make impossible to understand the meaning of the fields

Solutions:

- . provide a logical form layout
- . use form labels appropriately (<label>)

Labels

The <label> element is used to associate a text label to a form control. This allows a screen reader to read the associated label text when the user navigates to the form control.

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<label for="address">Enter your address:</label><br>
<textarea id="address" name="addresstext"></textarea>
```

Labels <http://webaim.org/techniques/forms/controls>

Examples of barriers for color blind people

Color coding

Barrier: if color alone is used to convey information, people who cannot differentiate between certain colors and users with devices that have non-color or non-visual displays will not receive the information

Solution: ensure that all information conveyed with color is also available without color, for example from context or markup (shape, position..)

Color combinations

Barrier: when foreground and background colors are too close to the same hue, they may not provide sufficient contrast when viewed using monochrome displays or by people with different types of color deficits

Solution: ensure that foreground and background color combinations provide sufficient contrast (use a color contrast checker)

Online color contrast checkers

checkmycolours.com contrastchecker.com

Examples of barriers for people with low vision

Horizontal scrolling

Barrier: users can increase font size using browser hot keys.. but the resulting page can be too large compared to the browser

Solutions:

- . use «fluid design» where text can wrap
- . use «responsive design» (to resize, hide, shrink, enlarge, or move the content to make it look good on any screen)

Responsive design

Responsive web sites are based on media queries, that allow the page to use different CSS style rules according to the characteristics of the device (e.g. width of the browser)

https://www.w3schools.com/css/css_rwd_mediaqueries.asp -

Pictures of text

Barrier: pictures of text become blurry when enlarged, take longer to download, and are inefficient for the website author to edit.

Solution: use text, not pictures of text.

You can attain very good visual effects using text and CSS.

Examples of barriers for deaf people

Sound in videos

Barrier: some important info may be conveyed through the sound of movies or animations.. useless for a deaf person

Solution: caption video and also provide a descriptive transcript.

Captioned video is accessible to people who can't hear it.

Captions make video full-text searchable.

Captions can be automatically translated to other languages during playback.

Captions

Captions are text versions of the spoken word presented

within multimedia.

Common web accessibility guidelines indicate that captions should be:

Synchronized -the text content should appear at approximately the same time that audio would be available

Equivalent -content provided in captions should be equivalent to that of the spoken word

Accessible -caption content should be readily accessible and available to those who need it

Captioning videos

There are many free, easy-to-use tools available that support the process of transcribing and captioning videos.

E.g. www.amara.org

Examples of barriers for motion impaired people

Bad use of Javascript

Often motion impaired people do not use the mouse (just like blind people)

Barrier: some JavaScript event handlers are device dependent (e.g., they require the use of a mouse)

Solutions:

- . make sure that your page does not rely on mouse to function
- . provide keyboard-based alternatives (onFocus instead of onMouseover)

Too many links

Often motion impaired people use the TAB key to move from link to link.

Barrier: long lists of links (or redundant links) tire the user and slow down browsing

Solution: provide a method that allows users to skip navigation or other elements that repeat on every page.

Skip... links

Allow users to skip repetitive elements on the page: this is usually accomplished by providing a "Skip to Content," "Skip to Main Content," or "Skip Navigation" link at the top of the page which jumps to the main content of the page.

Examples of barriers for people with cognitive disabilities

Memory; problem-solving; attention;
reading, linguistic, and verbal comprehension;
math comprehension; visual comprehension...

Cluttered layout

Barrier: complex relationships between parts of a page may be difficult for people with cognitive disabilities and people with visual disabilities to interpret.

Solutions:

- . simplify the layout!
- . divide large blocks of information into more manageable groups where natural and appropriate
- . standardize and be consistent
- . avoid too many choices
- . identify where on the site the user currently is.

Long texts

Dyslexia is a learning disability that impairs a person's fluency or accuracy in being able to read, write, and spell

Barrier: long blocks of unbroken paragraph text are not only hard for dyslexic users to read, but for non-dyslexic users too.

It's easy for dyslexic readers to lose their place with long paragraphs.

Solution: use short paragraphs that express one idea. (more breaks between ideas)

Complex texts

Barrier: a single means (text) can not always be sufficient to guarantee the understanding..

Solutions:

- . use the clearest and simplest language appropriate for a site's content.
- . supplement text with graphic or auditory presentations where they will facilitate comprehension of the page.

Use of graphics to supplement text

Use of graphics to substitute for text

Flashing visual content

Barrier: some people with seizure disorders (epilepsy)

can have a seizure triggered by flashing visual content
Solution: the WACG outline specific thresholds for size, frequency (faster than three flashes per second), intensity or contrast of the flashes, and red color; in general, if the content flashes more than three times per second, is notably large (a small animating image would not cause a seizure), has bright contrast in the flashes, it may cause a seizure and should be avoided.

Many other barriers

Many other situations require specific techniques to ensure accessibility

Web Accessibility Laws and standards

Rehabilitation Act and Section 508 (USA)

Rehabilitation Act: federal law (1973) that prohibits discrimination in the federal workplace against people with disabilities

Section 508: directives for removing barriers to the access to information and services in the field of IT

Section 508

Requires that when Federal agencies develop, procure, maintain, or use electronic and information technology, Federal employees with disabilities have access to and use of information and data that is comparable to the access and use by Federal employees who are not individuals with disabilities

Defines guidelines not only for the Web, but for electronic appliances in general as well

Became enforceable on June 21st, 2001

WAI: Web Accessibility Initiative (W3C)

WAI = Web Accessibility Initiative (1997)

develops guidelines widely regarded as the international standard for Web accessibility

creates support materials to help understand and implement Web accessibility

WCAG 1.0

WCAG 1.0 = Web Content Accessibility Guidelines
14 guidelines composed of checkpoints.

Became an official recommendation in 1999

<http://www.w3.org/TR/WCAG10/>

Several Web accessibility standards are based on the WCAG

WCAG 1.0 - Priority and Conformity

Every checkpoint has a priority level:

- . Priority 1: must be satisfied (otherwise it is impossible..)
- . Priority 2: should be satisfied (otherwise it is difficult..)
- . Priority 3: may be satisfied (otherwise it is not immediate..)

Conformity Level of a Web site:

- . Level A: Priority 1 checkpoints satisfied
- . Level AA: Priority 1 and Priority 2 checkpoints satisfied
- . Level AAA: Priority 1, Priority 2 and Priority 3 checkpoints satisfied

WCAG 1.0 - Example - Guideline 1

Provide equivalent alternatives to auditory and visual content

Provide content that, when presented to the user, conveys essentially

the same function or purpose as auditory or visual content

- . Checkpoint 1.1: Provide a text equivalent for every non-text element [Priority 1]
- . Checkpoint 1.2: Provide redundant text links for each active region of a server-side image map [Priority 1]
- . Checkpoint 1.3: Until user agents can automatically read aloud the text equivalent of a visual track, provide an auditory description of the important information of the visual track of a multimedia presentation [Priority 1]

WCAG 1.0 – Example – Guideline 2

Don't rely on color alone

Ensure that text and graphics are understandable when viewed without color.

- . Checkpoint 2.1: Ensure that all information conveyed with color is also available without color, for example from context or markup [Priority 1]
- . Checkpoint 2.2: Ensure that foreground and background color combinations provide sufficient contrast when viewed by someone having color deficits or when viewed on a black and white screen [Priority 2 for images, Priority 3 for text]

WCAG 1.0

Implementation techniques are also suggested

- : Provide equivalent alternatives to auditory and visual content
- : Don't rely on colour alone
- : Use markup and style sheets, and do so properly
- : Clarify natural language usage
- : Create tables that transform gracefully
- : Ensure that pages featuring new technologies transform gracefully
- : Ensure user control of time sensitive content changes
- : Ensure direct accessibility of embedded user interfaces
- : Design for device independence
- : User interim solutions
- : Use W3C technologies and guidelines
- : Provide context and orientation information
- : Provide clear navigation mechanisms
- : Ensure that documents are clear and simple

WCAG 2.0

WCAG 2.0 (December 2008)

<https://www.w3.org/TR/WCAG20/>

They are an evolution of the WCAG 1.0 (1999), that became a little bit obsolete due to technological advances. Despite this obsolescence, many laws still stand on these guidelines, so a new version was needed.

WCAG 2.0 organization

WCAG 2 are organized by 4 layers: principles, guidelines, success criteria and techniques to accomplish them.

. Principles provide the basis for Web accessibility: the website must be perceivable, operable, understandable, and robust.

- . 12 guidelines under the principles, similar to that on the WCAG 1, but with no levels of conformance.
- . Each guideline has a testable success criteria.
- . Each guideline and success criteria has its own informative techniques.

WCAG 2.0 principles

- . Principle 1: **Perceivable** - Information and user interface components must be presentable to users in ways they can perceive.
- . Principle 2: **Operable** - Users should be able to interact with the interface components and navigation elements.
- . Principle 3: **Understandable** - Information and the operation of user interface must be understandable.
- . Principle 4: **Robust** - Content must be robust enough that it can be interpreted reliably by a wide variety of user agents, including assistive technologies.

Accessibility in Europe (google i2010)

e-Europe 2002 action plan (June 2000)

"... Web sites of public organizations and institutions of member states and their contents must be defined so that they can be accessed by disabled people, allowing them to best exploit the opportunities offered by on-line administrations"

e-Europe 2005 plan: continuation of e-Europe 2002 several advanced services and initiatives for the creation of the "information society"

i2010 strategy: i2010 is the European Commission's new strategic framework laying out broad policy guidelines for the information society and the media.

Accessibility in Italy Accessibility in Italy

Legge Stanca -Promulgated on January 9th, 2004

Operative since August 8th, 2005

Article 1

Every person has the right to access all information sources and connected services; in particular, it is guaranteed the right of disabled persons to access electronic resources and computerized services of Civil Service

It envisages different accessibility levels and correspondent buttons..

Benefits of Web Accessibility

Investing in Accessibility

Like any new action within a company or organization, accessibility involves costs, but produces considerable benefits as well (in the middle term)

- . the number of site users grows
- . the site can be better navigated
- . several expenses decrease

Users growth 1/2

An accessible site is easier to find and use. This entails:

- . more users
- . more use by each user

In many cases, more users means financial benefits (more sales, more registrations, more funds, etc...)

When on-line services reduce personnel work (registrations, certificates, etc.), more use means more savings

Users growth 2/2

An accessible Web site is also usable by:

- . disabled persons
- . elders
- . people with low literacy
- . people not fluent in the language
- . people using old technologies
- . people using special browsers (e.g. small displays, b/w, ...)
- . injured, ill and elderly personnel within a company...

Savings / Costs

Savings

- . decrease of potential legal expenses
- . decrease of costs to create "alternative formats"
- . search engine optimization

Costs

- . personnel training
- . external consultancies
- . adjustment of company "rules"
- . evaluation and rethink of the current Web site